Supporting JFL learners with User Customized *Kanji* Learning System Using Computer Graphics

Noriko Uosaki
Center for International Education and Exchange, Osaka University
Osaka, Japan
uosaki@ciee.osaka-u.ac.jp

Kotaro Matsushita
Faculty of Informatics
Tokyo University of Information Sciences
Chiba, Japan
matusita@rsch.tuis.ac.jp

Hideo Suzuki
Faculty of Informatics
Tokyo University of Information Sciences
Chiba, Japan
suzuki@rsch.tuis.ac.jp

***Abstract*—This paper describes the development and evaluation of a user adaptive *kanji* learning system for JFL (Japanese as a foreign language) learners. Japanese language education for international students is an emergent issue in Japan. Especially learning *kanji* is a real challenge for them. Therefore we have tried to implement our newly developed *kanji* learning system, which received good reputations when introduced it into an elementary school, to a Japanese language class for non-native speakers. A pilot evaluation was conducted in order to find which aspects of our system to be modified to fit for JSL learners. The result of the pilot evaluation reveals that the participants' feedback was not good compared with that of elementary school children, especially about computer animations and fun factor. A user-unfriendliness about its navigation has been found. Our future works include the improvement of computer animations, the implementation of English navigation and the summary dashboards for reviewing.**

Keywords— JFL (Japanese as a Foreign Language); language learning; kanji; e-learning; learning system; computer graphics

I. INTRODUCTION

According to EF English Proficiency Index (3rd ed.) [1], Japan is considered as one of the countries where people have moderate English proficiency. However there are quite a few international students in Japan who report the inability of Japanese local people to communicate in English as one of their biggest culture shocks. Therefore in order to live in Japan, learning Japanese is a necessity. About 140,000 international students are now studying in Japan [2]. Besides, Japanese Government put forward a "Plan for 300,000 Exchange Students" as a policy measure to create *A Japan that is Open to the World* in 2008 [3]. Since then, Japanese language education for international students has become an emergent issue. But learning Japanese language, especially learning complex characters called *kanji* (Chinese characters such as 書,現,億) is a big challenge for the students from non-*kanji* using countries. How can we support *kanji* learning? Therefore our main objective in this study is to facilitate complicated *kanji* learning process with our developed system, which encourages students to learn *kanji* by taking automatically generated two-choice quizzes. The detailed introduction about the system is given in Section III. Our hypothetical question is: Does our system contribute to facilitation of *kanji* learning? In order to find an answer to this question, a pilot evaluation was conducted. The rest of the paper is organized as follows. First the backgrounds of our study are described in Section II. Our *kanji* learning system is introduced in Section III. Section IV includes a pilot evaluation we have conducted to investigate the effectiveness of our system. Finally, we give the conclusion and the future work of this study.

II. BACKGROUNDS

A. Heavy workload of Learning Kanji

Learning *kanji* is a real challenge for learners of Japanese as a second language, especially for those, who are from non-*kanji* using countries [4]. In fact, it is also a challenge for Chinese speakers because the most *kanji* in Japanese have plural readings which are totally unfamiliar to them [5]. *Kanji* mainly consists of two different readings called "*on-yomi*"(similar to Chinese reading, but different), and "*kun-yomi*"(Japanese original reading). [4] has pointed out that there are two factors which make *kanji* learning difficult:1) affective factor and 2) complexities of *kanji* themselves. JFL learners are overwhelmed by the facts that there are thousands of *kanji* to be learned and that there are different ways of pronunciation in each *kanji*, part of which are from their misunderstanding. These negative affirmation prevents necessary input for language learning [6]. In addition, to the learners from an alphabetic background, *kanji* looks like a combination of complex nonsense lines. When they start learning *kanji*, they are told to learn four things all at once: pronunciation, meaning, writing, and stroke order of *kanji*, heavy burden of *kanji* learning [4]. How can we support them? [7] and [8] have reported that quiz function was effective in language learning. One of the main functions of our developed system is to generate two-choice quizzes automatically. We expect our system could play an important role in facilitating complicated *kanji* learning process for JFL learners.

978-1-4799-9958-3/15 $31.00 © 2015 IEEE
DOI 10.1109/IIAI-AAI.2015.251

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B. *Technology Enhanced learning*

The progress of information and communication technology (ICT) has enabled us to carry out such learning environment as multimedia learning, technology-enhanced learning (TEL), computer-based instruction (CBI). Personal computers and internet devices have been introduced into schools to allow students to use PCs, tablets, or mobile devices on one-to-one or one-to-group basis. In accordance with its progress, computer-generated graphics are becoming increasingly available in school education. It has been reported that animated graphics were superior to static graphics [9], and learning performance using dynamic media was significantly higher than those of the static textbook lesson [10]. In our previous studies, using computer graphics, we have developed educational materials and a learning system to display e-learning contents [11][12]. The learning system displays a user adaptive learning contents to facilitate learning processes. Users can select e-learning contents that they would like to learn on this system to take automatically-generated quizzes for the selected contents. In order to make our developed learning system more adaptive for users for more effective learning, in this study, we used the Japanese *kanji* characters as e-learning content displayed on our computer system.

III. KANJI LEARNING SYSTEM

A. *System design*

The system is a client-server application, which runs on the web browser. It is programmed using HTML, JavaScript and PHP. The computer graphics animations were created using the POV-Ray software [13]. Figs. 1 and 2 show the system configuration and the screen structure.

Fig. 1. System Configuration

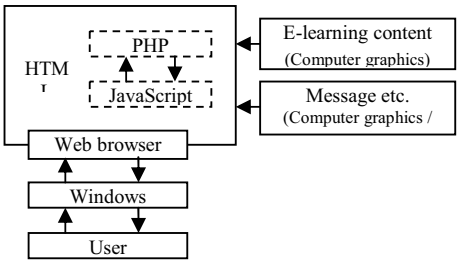
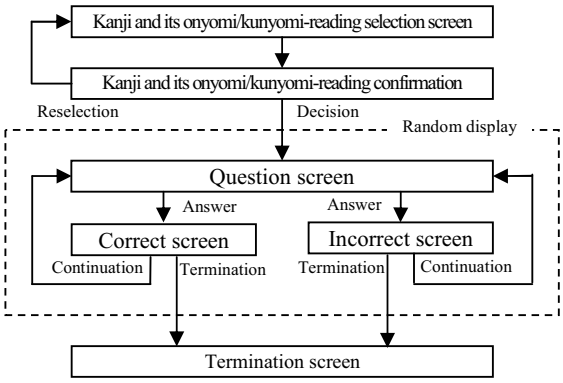


Fig. 2. Screen structure



B. *System Operation*

First the system displays the *kanji* and its *onyomi/kunyomi*-reading selection screen (Fig. 3). On this screen, learners select *kanji* characters and *onyomi* or *kunyomi* readings that they intend to learn. When they click the start button, two-choice quizzes are generated by the system (Fig. 4). They read a question and select the *kanji* which they think is correct. After they answers the question, the correct or incorrect answer screen shows the corresponding computer graphics animations (Fig. 5).

Fig. 3. *kanji* and *onyomi/kunyomi*-reading selection screen

Fig. 4. Two-choice Quiz interface

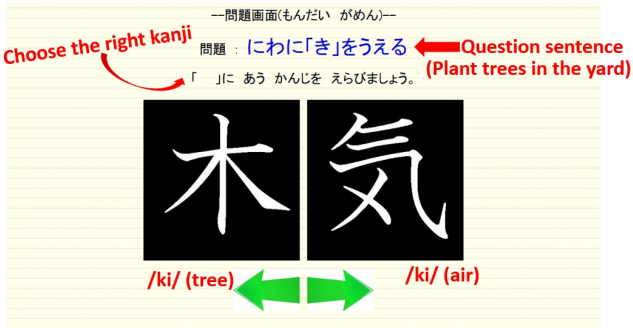
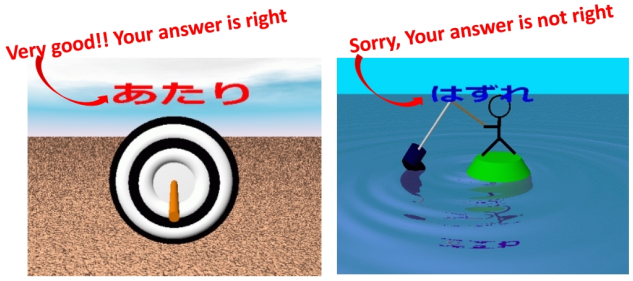


Fig. 5. Computer graphics animations of correct/wrong answer



(1) Correct (2) Wrong

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IV. PILOT EVALUATION

Our system was originally developed for the use of elementary school children [14]. This is our first trial to examine whether it is effective for JSL learners' kanji learning. Therefore this pilot evaluation was conducted in order to find which aspects of our system to be modified to fit for JSL learners.

A. Participants

Thirteen international students of CALL(computer assisted language learning) class which was held weekly at Osaka University participated in the evaluation (cf. TABLE 1). They all reported they had Internet-connected PCs at home and were mobile phone owners.

TABLE I. PARTICIPANTS

Student	*Nationality*	*Gender*	*Age*	*Length of stay in Japan*	*Major*
#1	Taiwan	f	25 - 29	6months	Economics
#2	Taiwan	f	25 - 29	6 months	Economic
#3	Denmark	f	20 -24	5 months	Engineering
#4	Netherlands	m	20 -24	6 months\	Engineering
#5	Netherlands	m	20 -24	6 months	Science
#6	USA	f	20 -24	11 months	Letters
#7	USA	m	20 -24	4 months	Engineering
#8	Australia	f	20 -24	6 months	Human Sciences
#9	Australia	m	20 -24	6 months	Engineering
#10	Germany	f	20 -24	5.5 Months	Human Sciences
#11	Germany	m	20 -24	1 year	Information Science and Technology
#12	Sweden	m	20 -24	1 year 3 months	Engineering science
#13	Sweden	m	25 - 29	6 months	Engineering science

B. Procedures

On the day the system was introduced, the participants received a briefing about how to use it such as how to select *onyomi/kunyomi*, how to select *kanji* and how to take quizzes during the class. They were assigned to use the system for their kanji learning as a home task. The evaluation was carried out over 3 weeks. After the evaluation phase, a questionnaire survey was conducted to obtain their feedback. Q1 and Q2 in the questionnaire were composed based on the technology acceptance model proposed by [15]. Q3 was created since a fun factor is considered to be important to enhance motivation for language learning [16]. Q4 and Q5 were created for examining the user acceptance of its interface and animation.

C. Results and Discussion

TABLE II shows the mean and SD (standard deviation) values of five-point-scale questionnaire. The highest score of 3.69 was given when asked whether it was easy to use the system, 3.38 when asked to rate how much they liked or disliked its interface, 3.31 when asked whether it was useful for kanji learning, 3.23 when asked whether it was fun to use the system, 2.92 when asked to rate how much they liked or disliked its animation. The participants' overall rating was not good as that of the fourth graders in our previous study, especially that of computer animations (2.92 Vs. 4.79) and fun factor (3.23 Vs. 4.60) [14]. I was found that there was a necessity to take some measures to add more fun to the system.

TABLE II. THE RESULT OF FIVE-POINT-SCALE QUESTIONNAIRE

	Question	*Mean*	*SD*
Q1	Was it helpful for your *kanji* learning?	3.31	0.88
Q2	Was it fun for you to use the system?	3.23	0.86
Q3	Was it easy for you to use *kanji* Learning System?	3.69	1.16
Q4	Please rate how much you liked or disliked its interface	3.38	0.47
Q5	Please rate how much you liked or disliked the animation?	2.92	1.03

TABLE III shows the result of the five-point-scale questionnaire of each participant. In order to find the answer to our hypothetical question: Does our system contribute to facilitation of *kanji* learning? Question 1 was asked. Seven out of 13 students gave 4, one student gave 5, two students gave 3, one gave 2 and two gave1. Therefore 8 out of 13 students seem to have derived some benefits from the system. However its mean value, 3.31 was not good as expected.

A positive correlation was detected only between *kanji* level and Q2. Since the instructions on the web interface were in Japanese, beginner level participants seem to have some difficulty in understanding its instructions (cf. Student #3's comment on Table IV). Though they received a briefing, English translation on the screen might have helped them.

TABLE III. CORRELATIONS BETWEEN KANJI-LEVEL AND RATING

Student	*kanji* - Level	Q1 Was it useful for your *kanji* learning ?	Q2 Was it easy for you to use *kanji* Learning System?	Q3 Was it fun for you to use the system? ?	Q4 Please rate how much you liked or disliked its interface	Q5 Please rate how much you liked or disliked the animation?
#1	6	4	5	4	4	4
#2	6	1	5	3	3	4
#3	1	2	2	2	3	3
#4	5.5	4	4	4	3	1
#5	1	4	3	3	4	4
#6	4	4	4	1	3	4
#7	2	4	3	4	3	3
#8	2	3	3	4	3	3
#9	1.5	3	3	3	3	2
#10	1	4	4	4	4	4
#11	2	1	5	3	4	3
#12	6	4	4	3	4	1
#13	1	5	3	4	3	2
r (correlation)		-0.05	**0.67**	-0.02	0.08	-0.16

Table IV shows free comments from the participants. As Student #3 pointed it out, there is a user-unfriendliness in the system since navigation of the site is in Japanese. Implementation of English navigation will be included in our future works. As for computer animations, most of their comments were negative ones. Students #7 reported that the

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animations took a bit of time to see if the answer is right or not. Student #13 reported that he felt they were annoying. Though it worked very well for elementary school children (mean: 4.79), university students had quite opposite opinions about that. Its refinement will be necessary. Students #7 also reported that it would be nice if there were a summary of the session after they finish. Implementation of a summary dashboard will also be included in our future works since it is a very important process in learning to review what they have learned [17].

TABLE IV. WHAT IS YOUR IMPRESSION OF USING KANJI LEARNING SYSTEM?

Student	*Comments*
#3	I had some problems using the *kanji* system because the navigation of the site is in Japanese. I don't always know what to do, and if I answered a question right or wrong. I would prefer to use my own English apps or wait till I get a better level of Japanese.
#7	The animations take a bit of time to see if you scored right or not. It'd also be nice if there were a summary of your session after you finish.
#9	Get rid of the animated images, Instead it should just state whether you are correct or not. It takes some time to see what the animated picture is meant to represent. Also if there was a way for the system to go back and review the *kanji* you got wrong, it would be a lot better. Because at the moment there is no way to review the ones you got wrong or that you should practice. It should make a note of this"
#12	Too long annoying animations, I want to study more effectively (see if right or wrong directly, rather with the right *kanji* in the example sentence to learn more). Would like to be able to use the keyboard arrows to select the left or right *kanji*. Would like to see total result when pressing done. Would like to be able to study the *kanji* I got wrong afterwards.
#13	It is a good idea but a big improvement for me would be a possibility of turning off the animations. There may be one, but I did not find it in that case.

V. CONCLUSIONS AND FUTURE WORKS

In this paper we have described the use of *Kanji* Learning System for facilitating international students to learn complicated Japanese *kanji*. Our hypothetical question was: Does our system contribute to facilitation of *kanji* learning? The result of the pilot evaluation reveals that the participants' feedback was not good when compared with that of elementary school children, especially about computer animations and fun factor. A user-unfriendliness about its navigation has been found. Our future works include the improvement of computer animations, the implementation of English navigation and the summary dashboards for reviewing. In order to add more fun to the system, we are going to introduce game elements in our system, such elements as in adventure games and RPG games. We expect that such game elements will be able to make it more enjoyable so that they can get more involved in *kanji* learning. With these refinements, more detail examination of effectiveness of our system will be conducted. We believe that our system will contribute to facilitation of *kanji* learning for JFL learners.

ACKNOWLEDGMENTS

Part of this research work was supported by the Grant-in-Aid for Scientific Research No.26350319 from the Ministry of Education, Culture, Sports, Science and Technology (MEXT) in Japan.

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